

Beta Distribution

The Beta distribution is the distribution most often used as the distribution of probabilities. In this section we are going to have a very meta discussion about how we represent probabilities. Until now probabilities have just been numbers in the range 0 to 1. However, if we have uncertainty about our probability, it would make sense to represent our probabilities as random variables (and thus articulate the relative likelihood of our belief).

Beta Random Variable

Notation: $X \sim \text{Beta}(a, b)$

Description: A belief distribution over the value of a probability p from a Binomial distribution after observing $a - 1$ successes and $b - 1$ fails.

Parameters: $a > 0$, the number successes + 1
 $b > 0$, the number of fails + 1

Support: $x \in [0, 1]$

PDF equation: $f(x) = B(a, b) \cdot x^{a-1} \cdot (1-x)^{b-1}$ where $B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$

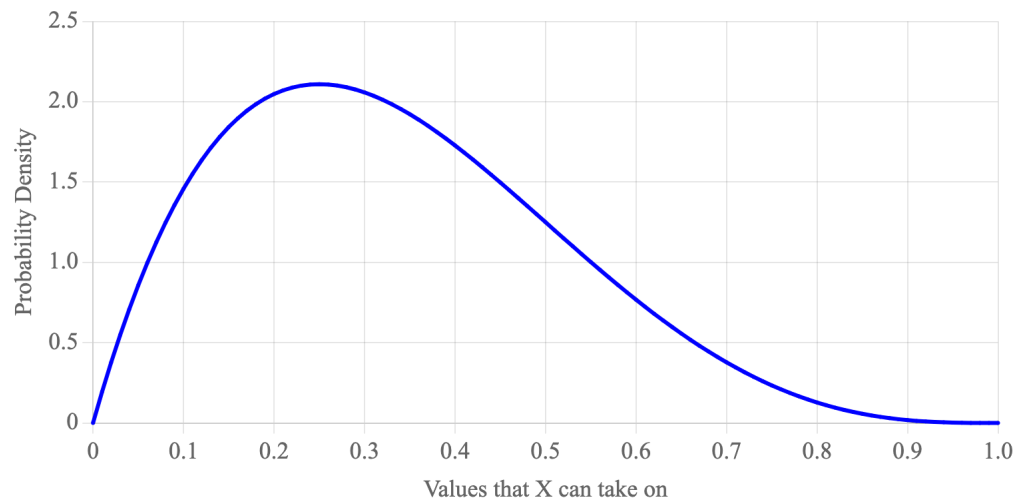
CDF equation: No closed form

Expectation: $E[X] = \frac{a}{a+b}$

Variance: $\text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}$

PDF graph:

Parameter a : Parameter b :



What is your Belief in p After 9 Heads in 10 Flips?

Imagine we have a coin and we would like to know its true probability of coming up heads, p . We flip the coin 10 times and observe 9 heads and 1 tail. What is your belief in p based off this evidence? Using the definition of probability we could guess that $p \approx \frac{9}{10}$. That number is a very rough estimate, especially since it is only based off 10 coin flips. Moreover the "point-value" $\frac{9}{10}$ does not have the ability to articulate how uncertain it is.

Could we instead have a random variable for the true probability? Formally, let X represent the true probability of the coin coming up heads. We don't use the symbol P for random variables, so X will have to do. If $X = 0.7$ then the probability of heads is 0.7. X must be a continuous random variable with support $[0, 1]$ since probabilities are continuous values which must be between 0 and 1.

Before flipping the coin, we could say that our belief about the coin's heads probability is uniform: $X \sim \text{Uni}(0, 1)$. Let H be a random variable for the number of heads and let T be a random variable for the number of tails observed. What is $P(X = x | H = 9, T = 1)$?

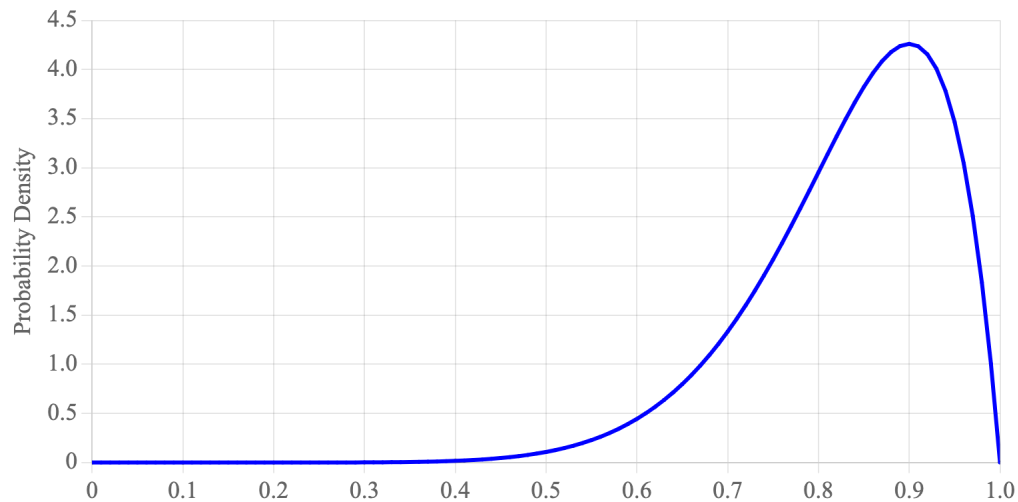
That probability is hard to think about! However it is much easier to reason about the probability with the condition reversed: $P(H = 9, T = 1 | X = x)$. This term asks the question: what is the probability of seeing 9 heads and 1 tail in 10 coin flips, given that the true probability of a heads is x . Convince yourself that this probability is just a binomial probability mass function with $n = 10$ experiments, and $p = x$ evaluated at $k = 9$ heads:

$$P(H = 9, T = 1 | X = x) = \binom{10}{9} x^9 (1 - x)^1$$

We are presented with a perfect context for [Bayes' theorem with random variables](#). We know a conditional probability in one direction and we would like to know it in the other:

$$\begin{aligned} f(X = x | H = 9, T = 1) &= \frac{P(H = 9, T = 1 | X = x) \cdot f(X = x)}{P(H = 9, T = 1)} && \text{Bayes Theorem} \\ &= \frac{\binom{10}{9} x^9 (1 - x)^1 \cdot f(X = x)}{P(H = 9, T = 1)} && \text{Binomial PMF} \\ &= \frac{\binom{10}{9} x^9 (1 - x)^1 \cdot 1}{P(H = 9, T = 1)} && \text{Uniform PDF} \\ &= \frac{\binom{10}{9}}{P(H = 9, T = 1)} x^9 (1 - x)^1 && \text{Constants to front} \\ &= K \cdot x^9 (1 - x)^1 && \text{Rename constant} \end{aligned}$$

Lets take a look at that function. For now we can let $K = 110$. Regardless of K we will get the same shape, just scaled:



What a beautiful image. It tells us relatively likelihood over the probability that is governing our coinflips. Here are a few observations from this chart:

1. Even after only 10 coin flips we are very confident that the true probability is > 0.5
2. It is almost 10 times more likely that $X = 0.9$ as it is that $X = 0.6$.
3. $f(X = 1) = 0$, which makes sense. How could we have flipped that one tail if the probability of heads was 1?

Wait but why?

In the derivation above for $f(X = x | H = 9, T = 1)$ we made the claim that $P(H = 9, T = 1)$ is a constant. A lot of folks find that hard to believe. Why is that the case?

It may be helpful to juxtapose $P(H = 9, T = 1)$ with $P(H = 9, T = 1 | X = x)$. The latter says "what is the probability of 9 heads, given the true probability is x ". The former says "what is the probability of 9 heads, under all possible assignments of x ". If you wanted to calculate $P(H = 9, T = 1)$ you could use the law of total probability:

$$\begin{aligned} P(H = 9, T = 1) \\ = \int_{y=0}^1 P(H = 9, T = 1 | X = y) f(X = y) \end{aligned}$$

That is a hard number to calculate, but it is in fact a constant with respect to x .

Beta Derivation

Let's generalize the derivation from the previous section, using h for the number of observed heads and t the number of observed tails.

If we let $H = h$ be the event that we saw h heads, and let $T = t$ be the event that we saw t tails in $h + t$ coinflips. We want to calculate the probability density function $f(X = x | H = h, T = t)$. We can use the exact same series of steps, starting with Bayes Theorem:

$$\begin{aligned} f(X = x | H = h, T = t) &= \frac{P(H = h, T = t | X = x) f(X = x)}{P(H = h, T = t)} && \text{Bayes Theorem} \\ &= \frac{\binom{h+t}{h} x^h (1-x)^t}{P(H = h, T = t)} && \text{Binomial PMF, Uniform PDF} \\ &= \frac{\binom{h+t}{h}}{P(H = h, T = t)} x^h (1-x)^t && \text{Moving terms around} \\ &= \frac{1}{c} \cdot x^h (1-x)^t && \text{where } c = \int_0^1 x^h (1-x)^t dx \end{aligned}$$

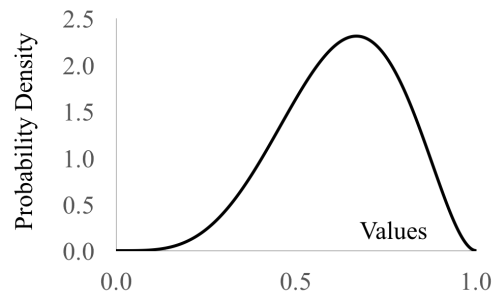
The equation that we arrived at when using a Bayesian approach to estimating our probability defines a probability density function and thus a random variable. The random variable is called a Beta distribution, and it is defined as follows:

The Probability Density Function (PDF) for $X \sim \text{Beta}(a, b)$ is:

$$f(X = x) = \begin{cases} \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1} & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{where } B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$$

A Beta distribution has $E[X] = \frac{a}{a+b}$ and $\text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}$. All modern programming languages have a package for calculating Beta CDFs. You will not be expected to compute the CDF by hand in CS109.

To model our estimate of the probability of a coin coming up heads: set $a = h + 1$ and $b = t + 1$. Beta is used as a random variable to represent a belief distribution of probabilities in contexts beyond estimating coin flips. For example perhaps a drug has been given to 6 patients, 4 of whom have been cured. We could express our belief in the probability that the drug can cure patients as $X \sim \text{Beta}(a = 5, b = 3)$:



Notice how the most likely belief for the probability of curing a patient, is $4/6$, the fraction of patients cured. This distribution shows that we hold a non-zero belief that the probability could be something other than $4/6$. It is unlikely that the probability is 0.01 or 0.09, but reasonably likely that it could be 0.5.

Beta as a Prior

You can set $X \sim \text{Beta}(a, b)$ as a prior to reflect how biased you think the coin is apriori to flipping it. This is a subjective judgment that represent $a + b - 2$ "imaginary" trials with $a - 1$ heads and $b - 1$ tails. If you then observe $h + t$ real trials with h heads you can update your belief. Your new belief would be, $X \sim \text{Beta}(a + h, b + t)$. Using the prior $\text{Beta}(1, 1) = \text{Uni}(0, 1)$ is the same as saying we haven't seen any "imaginary" trials, so apriori we know nothing about the coin. Here is the proof for the distribution of X when the prior was a Beta too:

If our prior belief is $X \sim \text{Beta}(a, b)$, then our posterior is $\text{Beta}(a + h, b + t)$:

$$\begin{aligned}
 f(X = x | H = h, T = t) &= \frac{P(H = h, T = t | X = x) f(X = x)}{P(H = h, T = t)} && \text{Bayes Theorem} \\
 &= \frac{\binom{h+t}{h} x^h (1-x)^t \cdot \frac{1}{c} \cdot x^{a-1} (1-x)^{b-1}}{P(H = h, T = t)} && \text{Beta PMF, Uniform PDF} \\
 &= K \cdot x^h (1-x)^t \cdot x^{a-1} (1-x)^{b-1} && \text{Combine Constants} \\
 &= K \cdot x^{a+h-1} (1-x)^{b+t-1} && \text{Combine Like Bases}
 \end{aligned}$$

Which is the PDF of $\text{Beta}(a + h, b + t)$

It is pretty convenient that if we have a Beta prior belief, then our posterior belief is also Beta. This makes Betas especially convenient to work with, in code and in proof, if there are many updates that you will make to your belief over time. This property where the type of distribution is the same before and after an observation is called a conjugate prior.

Quick question: Are you allowed to just make up priors and imaginary trials? Some folks think that is fine (they are called Bayesians) and some folks think that you shouldn't make up prior beliefs (they are called frequentists). In general, for small data it can make you much better at making predictions if you are able to come up with a good prior belief.

Observation: There is a deep connection between the beta-prior and the uniform-prior (which we used initially). It turns out that $\text{Beta}(1, 1) = \text{Uni}(0, 1)$. Recall that $\text{Beta}(1, 1)$ means 0 imaginary heads and 0 imaginary tails.

Adding Random Variables

In this section on uncertainty theory we are going to explore some of the great results in probability theory. As a gentle introduction we are going to start with convolution. Convolution is a very fancy way of saying "adding" two different random variables together. The name comes from the fact that adding two random variables requires you to "convolve" their distribution functions. It is interesting to study in detail because (1) many natural processes can be modelled as the sum of random variables, and (2) because mathematicians have made great progress on proving convolution theorems. For some particular random variables computing convolution has closed form equations. Importantly convolution is the sum of the random variables themselves, not the addition of the probability density functions (PDF)s that correspond to the random variables.

1. [Adding Two Random Variables](#)
2. [Sum of Independent Poissons](#)
3. [Sum of Independent Binomials](#)
4. [Sum of Independent Normals](#)
5. [Sum of Independent Uniforms](#)

Adding Two Random Variables

Deriving an expression for the likelihood for the sum of two random variables requires an interesting insight. If your random variables are discrete then the probability that $X + Y = n$ is the sum of mutually exclusive cases where X takes on a values in the range $[0, n]$ and Y takes on a value that allows the two to sum to n . Here are a few examples $X = 0$ and $Y = n$, $X = 1$ and $Y = n - 1$ etc. In fact all of the mutually exclusive cases can be enumerated in a sum:

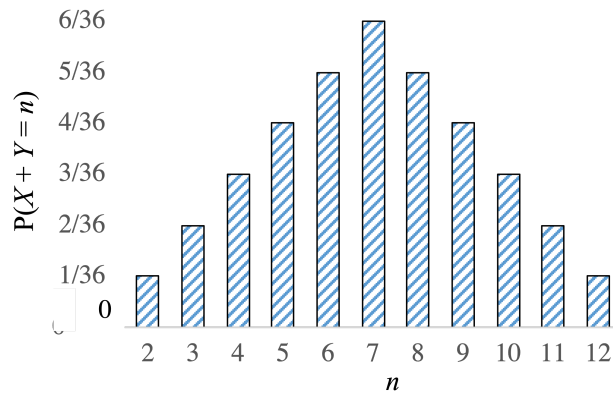
Def: General Rule for the Convolution of Discrete Variables

$$P(X + Y = n) = \sum_{i=-\infty}^{\infty} P(X = i, Y = n - i)$$

If the random variables are independent you can further decompose the term $P(X = i, Y = n - i)$. Let's expand on some of the mutually exclusive cases where $X + Y = n$:

i	X	Y	
0	0	n	$P(X = 0, Y = n)$
1	1	$n - 1$	$P(X = 1, Y = n - 1)$
2	2	$n - 2$	$P(X = 2, Y = n - 2)$
...			
n	n	0	$P(X = n, Y = 0)$

Consider the sum of two independent dice. Let X and Y be the outcome of each dice. Here is the probability mass function for the sum $X + Y$:



Let's use this context to practice deriving the sum of two variables, in this case $P(X + Y = n)$, starting with the General Rule for the Convolution of Discrete Random Variables. We start by considering values of n between 2 and 7. In this range $P(X = i, Y = n - i) = \frac{1}{36}$ for all values of i between 1 and $n - 1$. There is exactly one outcome of the two die where $X = i$ and $Y = n - i$. For values of i outside this range $n - i$ is not a valid dice outcome and $P(X = i, Y = n - i) = 0$:

$$\begin{aligned}
 P(X + Y = n) &= \sum_{i=-\infty}^{\infty} P(X = i, Y = n - i) \\
 &= \sum_{i=1}^{n-1} P(X = i, Y = n - i) \\
 &= \sum_{i=1}^{n-1} \frac{1}{36} \\
 &= \frac{n-1}{36}
 \end{aligned}$$

For values of n greater than 7 we could use the same approach, though different values of i would make $P(X = i, Y = n - i)$ non-zero.

This derivation for a general rule has a continuous equivalent:

$$f(X + Y = n) = \int_{i=-\infty}^{\infty} f(X = n - i, Y = i) di$$

Sum of Independent Poissons

For any two Poisson random variables: $X \sim \text{Poi}(\lambda_1)$ and $Y \sim \text{Poi}(\lambda_2)$ the sum of those two random variables is another Poisson: $X + Y \sim \text{Poi}(\lambda_1 + \lambda_2)$. This holds even when λ_1 is not the same as λ_2 .

How could we prove the above claim?

Example derivation:

Let's go about proving that the sum of two independent Poisson random variables is also Poisson. Let $X \sim \text{Poi}(\lambda_1)$ and $Y \sim \text{Poi}(\lambda_2)$ be two independent random variables, and $Z = X + Y$. What is $P(Z = n)$?

$$\begin{aligned}
P(Z = n) &= P(X + Y = n) \\
&= \sum_{k=-\infty}^{\infty} P(X = k, Y = n - k) && \text{(Convolution)} \\
&= \sum_{k=-\infty}^{\infty} P(X = k)P(Y = n - k) && \text{(Independence)} \\
&= \sum_{k=0}^n P(X = k)P(Y = n - k) && \text{(Range of } X \text{ and } Y) \\
&= \sum_{k=0}^n e^{-\lambda_1} \frac{\lambda_1^k}{k!} e^{-\lambda_2} \frac{\lambda_2^{n-k}}{(n-k)!} && \text{(Poisson PMF)} \\
&= e^{-(\lambda_1 + \lambda_2)} \sum_{k=0}^n \frac{\lambda_1^k \lambda_2^{n-k}}{k!(n-k)!} \\
&= \frac{e^{-(\lambda_1 + \lambda_2)}}{n!} \sum_{k=0}^n \frac{n!}{k!(n-k)!} \lambda_1^k \lambda_2^{n-k} \\
&= \frac{e^{-(\lambda_1 + \lambda_2)}}{n!} (\lambda_1 + \lambda_2)^n && \text{(Binomial theorem)}
\end{aligned}$$

Note that the Binomial Theorem (which we did not cover in this class, but is often used in contexts like expanding polynomials) says that for two numbers a and b and positive integer n , $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$.

Sum of Independent Binomials with equal p

For any two independent Binomial random variables with the same "success" probability p :
 $X \sim \text{Bin}(n_1, p)$ and $Y \sim \text{Bin}(n_2, p)$ the sum of those two random variables is another binomial:
 $X + Y \sim \text{Bin}(n_1 + n_2, p)$.

This result hopefully makes sense. The convolution is the number of successes across X and Y . Since each trial has the same probability of success, and there are now $n_1 + n_2$ trials, which are all independent, the convolution is simply a new Binomial. This rule does not hold when the two Binomial random variables have different parameters p .

Sum of Independent Normals

For any two independent normal random variables $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ the sum of those two random variables is another normal: $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Again this only holds when the two normals are independent.

Sum of Independent Uniforms

If X and Y are independent uniform random variables where $X \sim \text{Uni}(0, 1)$ and $Y \sim \text{Uni}(0, 1)$:

$$f(X + Y = n) = \begin{cases} n & \text{if } 0 < n \leq 1 \\ 2 - n & \text{if } 1 < n \leq 2 \\ 0 & \text{else} \end{cases}$$

Example derivation:

Calculate the PDF of $X + Y$ for independent uniform random variables $X \sim \text{Uni}(0, 1)$ and $Y \sim \text{Uni}(0, 1)$? First plug in the equation for general convolution of independent random variables:

$$\begin{aligned}
f(X + Y = n) &= \int_{i=0}^1 f(X = n - i, Y = i) di \\
&= \int_{i=0}^1 f(X = n - i) f(Y = i) di && \text{Independence} \\
&= \int_{i=0}^1 f(X = n - i) di && \text{Because } f(Y = y) = 1
\end{aligned}$$

It turns out that is not the easiest thing to integrate. By trying a few different values of n in the range $[0, 2]$ we can observe that the PDF we are trying to calculate is discontinuous at the point $n = 1$ and thus will be easier to think about as two cases: $n < 1$ and $n > 1$. If we calculate $f(X + Y = n)$ for both cases and correctly constrain the bounds of the integral we get simple closed forms for each case:

$$f(X + Y = n) = \begin{cases} n & \text{if } 0 < n \leq 1 \\ 2 - n & \text{if } 1 < n \leq 2 \\ 0 & \text{else} \end{cases}$$

Central Limit Theorem

There are two ways that you could state the central limit theorem. Either that the sum of IID random variables is normally distributed, or that the average of IID random variables is normally distributed.

The Central Limit Theorem (Sum Version)

Let $X_1, X_2 \dots X_n$ be independent and identically distributed random variables. The **sum** of these random variables approaches a normal as $n \rightarrow \infty$:

$$\sum_{i=1}^n X_i \sim N(n \cdot \mu, n \cdot \sigma^2)$$

Where $\mu = E[X_i]$ and $\sigma^2 = \text{Var}(X_i)$. Note that since each X_i is identically distributed they share the same expectation and variance.

At this point you probably think that the central limit theorem is awesome. But it gets even better. With some algebraic manipulation we can show that if the sample mean of IID random variables is normal, it follows that the sum of equally weighted IID random variables must also be normal:

The Central Limit Theorem (Average Version)

Let $X_1, X_2 \dots X_n$ be independent and identically distributed random variables. The **average** of these random variables approaches a normal as $n \rightarrow \infty$:

$$\frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Where $\mu = E[X_i]$ and $\sigma^2 = \text{Var}(X_i)$.

Central Limit Theorem Intuition

In the previous section we explored what happens when you [add two random variables](#). What happens when you add more than two random variables? For example, what if I wanted to add up 100 different uniform random variables:

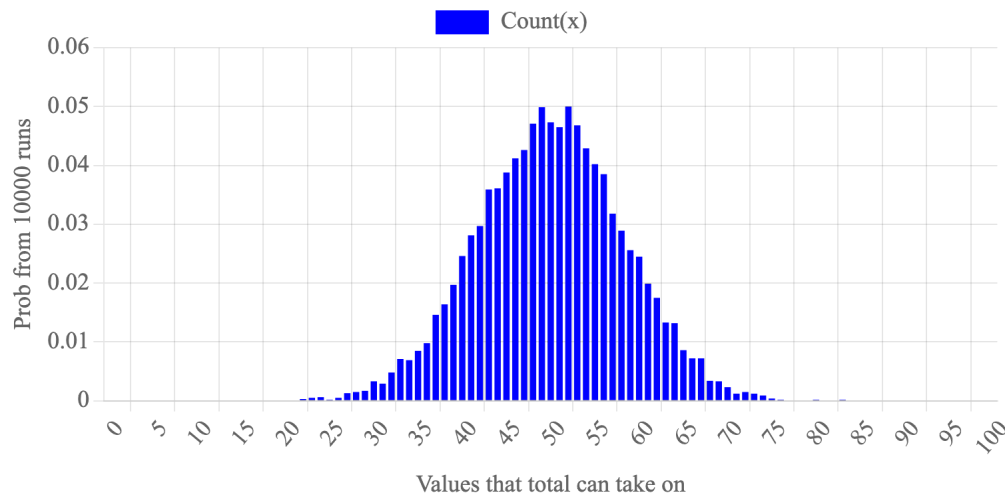
```
from random import random

def add_100_uniforms():
    total = 0
    for i in range(100):
        # returns a sample from uniform(0, 1)
        x_i = random()
        total += x_i
    return total
```

The value, **total** returned by this function will be a random variable. Hit the button below to run the function and observe the resulting value of total:

```
add_100_uniforms()    total: 47.00654
```

What does total look like as a distribution? Let's calculate **total** many times and visualize the histogram of values it produces.



10,000 more runs

That is interesting! **total** which is the sum of 100 independent uniforms looks normal. Is that a special property of uniforms? No! It turns out to work for almost any type of distribution (as long as the thing you are adding has finite mean and finite variance, everything we have covered in this reader).

- Sum of 40 X_i where $X_i \sim \text{Beta}(a = 5, b = 4)$? Normal.
- Sum of 90 X_i where $X_i \sim \text{Poi}(\lambda = 4)$? Normal.
- Sum of 50 dice-rolls? Normal.
- Average of 10000 X_i where $X_i \sim \text{Exp}(\lambda = 8)$? Normal.

For any distribution the sum, or average, of n independent equally-weighted samples from that distribution, will be normal.

Continuity Correction

Now we can see that the Binomial Approximation using a Normal actually derives from the central limit theorem. Recall that, when computing probabilities for a normal approximation, we had to to use a [continuity correction](#). This was because we were approximating a discrete random variable (a binomial) with a continuous one (a normal). You should use a continuity correction any time your normal is approximating a discrete random variable. The rules for a general continuity correction are the same as the rules for the binomial-approximation continuity correction.

In the motivating example above, where we added 100 uniforms, a continuity correction isn't needed because the sum of uniforms is continuous. In the dice sum example below, a continuity correction is needed because die outcomes are discrete.

Examples

Example:

You will roll a 6 sided dice 10 times. Let X be the total value of all 10 dice $= X_1 + X_2 + \dots + X_{10}$. You win the game if $X \leq 25$ or $X \geq 45$. Use the central limit theorem to calculate the probability that you win. Recall that $E[X_i] = 3.5$ and $\text{Var}(X_i) = \frac{35}{12}$.

Let Y be the approximating normal. By the Central Limit Theorem $Y \sim N(10 \cdot E[X_i], 10 \cdot \text{Var}(X_i))$. Substituting in the known values for expectation and variance: $Y \sim N(35, 29.2)$

$$\begin{aligned}
& P(X \leq 25 \text{ or } X \geq 45) \\
&= P(X \leq 25) + P(X \geq 45) \\
&\approx P(Y < 25.5) + P(Y > 44.5) && \text{Continuity Correction} \\
&\approx P(Y < 25.5) + [1 - P(Y < 44.5)] \\
&\approx \Phi\left(\frac{25.5 - 35}{\sqrt{29.2}}\right) + \left[1 - \Phi\left(\frac{44.5 - 35}{\sqrt{29.2}}\right)\right] && \text{Normal CDF} \\
&\approx \Phi(-1.76) + [1 - \Phi(1.76)] \\
&\approx 0.039 + (1 - 0.961) \approx 0.078
\end{aligned}$$

Example:

Say you have a new algorithm and you want to test its running time. You have an idea of the variance of the algorithm's run time: $\sigma^2 = 4\text{sec}^2$ but you want to estimate the mean: $\mu = t\text{sec}$. You can run the algorithm repeatedly (IID trials). How many trials do you have to run so that your estimated runtime = $t \pm 0.5$ with 95% certainty? Let X_i be the run time of the i -th run (for $1 \leq i \leq n$).

$$0.95 = P(-0.5 \leq \frac{\sum_{i=1}^n X_i}{n} - t \leq 0.5)$$

By the central limit theorem, the standard normal Z must be equal to:

$$\begin{aligned}
Z &= \frac{(\sum_{i=1}^n X_i) - n\mu}{\sigma\sqrt{n}} \\
&= \frac{(\sum_{i=1}^n X_i) - nt}{2\sqrt{n}}
\end{aligned}$$

Now we rewrite our probability inequality so that the central term is Z :

$$\begin{aligned}
0.95 &= P(-0.5 \leq \frac{\sum_{i=1}^n X_i}{n} - t \leq 0.5) = P(\frac{-0.5\sqrt{n}}{2} \leq \frac{\sum_{i=1}^n X_i}{n} - t \leq \frac{0.5\sqrt{n}}{2}) \\
&= P(\frac{-0.5\sqrt{n}}{2} \leq \frac{\sqrt{n}}{2} \frac{\sum_{i=1}^n X_i}{n} - \frac{\sqrt{n}}{2} t \leq \frac{0.5\sqrt{n}}{2}) = P(\frac{-0.5\sqrt{n}}{2} \leq \frac{\sum_{i=1}^n X_i}{2\sqrt{n}} - \frac{\sqrt{n}}{\sqrt{n}} \frac{\sqrt{n}t}{2} \leq) \\
&= P(\frac{-0.5\sqrt{n}}{2} \leq \frac{\sum_{i=1}^n X_i - nt}{2\sqrt{n}} \leq \frac{0.5\sqrt{n}}{2}) \\
&= P(\frac{-0.5\sqrt{n}}{2} \leq Z \leq \frac{0.5\sqrt{n}}{2})
\end{aligned}$$

And now we can find the value of n that makes this equation hold.

$$\begin{aligned}
0.95 &= \phi\left(\frac{\sqrt{n}}{4}\right) - \phi\left(-\frac{\sqrt{n}}{4}\right) = \phi\left(\frac{\sqrt{n}}{4}\right) - (1 - \phi\left(\frac{\sqrt{n}}{4}\right)) \\
&= 2\phi\left(\frac{\sqrt{n}}{4}\right) - 1 \\
0.975 &= \phi\left(\frac{\sqrt{n}}{4}\right) \\
\phi^{-1}(0.975) &= \frac{\sqrt{n}}{4} \\
1.96 &= \frac{\sqrt{n}}{4} \\
n &= 61.4
\end{aligned}$$

Thus it takes 62 runs. If you are interested in how this extends to cases where the variance is unknown, look into variations of the students' t-test.

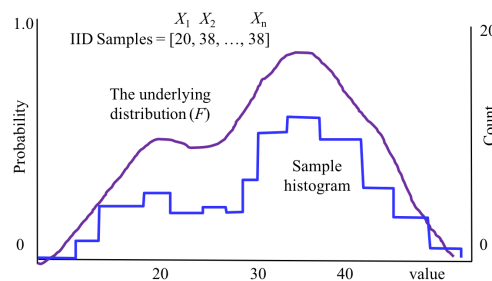
Sampling

In this section we are going to talk about statistics calculated on samples from a population. We are then going to talk about probability claims that we can make with respect to the original population -- a central requirement for most scientific disciplines.

Let's say you are the king of Bhutan and you want to know the average happiness of the people in your country. You can't ask every single person, but you could ask a random subsample. In this next section we will consider principled claims that you can make based on a subsample. Assume we randomly sample 200 Bhutanese and ask them about their happiness. Our data looks like this: 72, 85, ..., 71. You can also think of it as a collection of $n = 200$ I.I.D. (independent, identically distributed) random variables $X_1, X_2, \dots, X_n$.

Understanding Samples

The idea behind sampling is simple, but the details and the mathematical notation can be complicated. Here is a picture to show you all of the ideas involved:



The theory is that there is some large population (for example the 774,000 people who live in Bhutan). We collect a sample of n people at random, where each person in the population is equally likely to be in our sample. From each person we record one number (for example their reported happiness). We are going to call the number from the i th person we sampled X_i . One way to visualize your samples $X_1, X_2, \dots, X_n$ is to make a histogram of their values.

We make the assumption that all of our X_i s are identically distributed. That means that we are assuming there is a single underlying distribution F that we drew our samples from. Recall that a distribution for discrete random variables should define a probability mass function.

Estimating Mean and Variance from Samples

We assume that the data we look at are IID from the same underlying distribution (F) with a true mean (μ) and a true variance (σ^2). Since we can't talk to everyone in Bhutan we have to rely on our sample to estimate the mean and variance. From our sample we can calculate a sample mean ($\bar{X}$) and a sample variance (S^2). These are the best guesses that we can make about the true mean and true variance.

$$\bar{X} = \sum_{i=1}^n \frac{X_i}{n}$$
$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

The first question to ask is, are those unbiased estimates? Yes. Unbiased, means that if we were to repeat this sampling process many times, the expected value of our estimates should be equal to the true values we are trying to estimate. We will prove that that is the case for $\bar{X}$. The proof for S^2 is in lecture slides.

$$\begin{aligned}
E[\bar{X}] &= E\left[\sum_{i=1}^n \frac{X_i}{n}\right] = \frac{1}{n} E\left[\sum_{i=1}^n X_i\right] \\
&= \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} n\mu = \mu
\end{aligned}$$

The equation for sample mean seems related to our understanding of expectation. The same could be said about sample variance except for the surprising $(n - 1)$ in the denominator of the equation. Why $(n - 1)$? That denominator is necessary to make sure that the $E[S^2] = \sigma^2$.

The intuition behind the proof is that sample variance calculates the distance of each sample to the sample mean, **not** the true mean. The sample mean itself varies, and we can show that its variance is also related to the true variance.

Standard Error

Ok, you convinced me that our estimates for mean and variance are not biased. But now I want to know how much my sample mean might vary relative to the true mean.

$$\begin{aligned}
\text{Var}(\bar{X}) &= \text{Var}\left(\sum_{i=1}^n \frac{X_i}{n}\right) = \left(\frac{1}{n}\right)^2 \text{Var}\left(\sum_{i=1}^n X_i\right) \\
&= \left(\frac{1}{n}\right)^2 \sum_{i=1}^n \text{Var}(X_i) = \left(\frac{1}{n}\right)^2 \sum_{i=1}^n \sigma^2 = \left(\frac{1}{n}\right)^2 n\sigma^2 = \frac{\sigma^2}{n} \\
&\approx \frac{S^2}{n} \\
\text{Std}(\bar{X}) &\approx \sqrt{\frac{S^2}{n}}
\end{aligned}$$

That term, $\text{Std}(\bar{X})$, has a special name. It is called the standard error and its how you report uncertainty of estimates of means in scientific papers (and how you get error bars). Great! Now we can compute all these wonderful statistics for the Bhutanese people. But wait! You never told me how to calculate the $\text{Std}(S^2)$. That is hard because the central limit theorem doesn't apply to the computation of S^2 . Instead we will need a more general technique. See the next chapter: [Bootstrapping](#)

Let's say we calculate the our sample of happiness has $n = 200$ people. The sample mean is $\bar{X} = 83$ (what is the unit here? happiness score?) and the sample variance is $S^2 = 450$. We can now calculate the standard error of our estimate of the mean to be 1.5. When we report our results we will say that our estimate of the average happiness score in Bhutan is 83 ± 1.5 . Our estimate of the variance of happiness is $450 \pm ?$.

Bootstrapping

The bootstrap is a newly invented statistical technique for both understanding distributions of statistics and for calculating p -values (a p -value is the probability that a scientific claim is incorrect). It was invented here at Stanford in 1979 when mathematicians were just starting to understand how computers, and computer simulations, could be used to better understand probabilities.

The first key insight is that: if we had access to the underlying distribution (F) then answering almost any question we might have as to how accurate our statistics are becomes straightforward. For example, in the previous section we gave a formula for how you could calculate the sample variance from a sample of size n . We know that in expectation our sample variance is equal to the true variance. But what if we want to know the probability that the true variance is within a certain range of the number we calculated? That question might sound dry, but it is critical to evaluating scientific claims! If you knew the underlying distribution, F , you could simply repeat the experiment of drawing a sample of size n from F , calculate the sample variance from our new sample and test what portion fell within a certain range.

The next insight behind bootstrapping is that the best estimate that we can get for F is from our sample itself! The simplest way to estimate F (and the one we will use in this class) is to assume that the $P(X = k)$ is simply the fraction of times that k showed up in the sample. Note that this defines the probability mass function of our estimate $\hat{F}$ of F .

```
def bootstrap(sample):
    N = number of elements in sample
    pmf = estimate the underlying pmf from the sample
    stats = []
    repeat 10,000 times:
        resample = draw N new samples from the pmf
        stat = calculate your stat on the resample
        stats.append(stat)
    stats can now be used to estimate the distribution of the stat
```

Bootstrapping is a reasonable thing to do because the sample you have is the best and only information you have about what the underlying population distribution actually looks like. Moreover most samples will, if they're randomly chosen, look quite like the population they came from.

To calculate $\text{Var}(S^2)$ we could calculate S_i^2 for each resample i and after 10,000 iterations, we could calculate the sample variance of all the S_i^2 s. You might be wondering why the resample is the same size as the original sample (n). The answer is that the variation of the variation of stat that you are calculating could depend on the size of the sample (or the resample). To accurately estimate the distribution of the stat we must use resamples of the same size.

The bootstrap has strong theoretic grantees, and is accepted by the scientific community. It breaks down when the underlying distribution has a "long tail" or if the samples are not I.I.D.

Example of p-value calculation

We are trying to figure out if people are happier in Bhutan or in Nepal. We sample $n_1 = 200$ individuals in Bhutan and $n_2 = 300$ individuals in Nepal and ask them to rate their happiness on a scale from 1 to 10. We measure the sample means for the two samples and observe that people in Nepal are slightly happier--the difference between the Nepal sample mean and the Bhutan sample mean is 0.5 points on the happiness scale.

If you want to make this claim scientific you should calculate a p -value. A p -value is the probability that, when the null hypothesis is true, the statistic measured would be equal to, or more extreme than, than the value you are reporting. The null hypothesis is the hypothesis that there is no relationship between two measured phenomena or no difference between two groups.

In the case of comparing Nepal to Bhutan, the null hypothesis is that there is no difference between the distribution of happiness in Bhutan and Nepal. The null hypothesis argument is: there is no difference in the distribution of happiness between Nepal and Bhutan. When you drew samples, Nepal had a mean that 0.5 points larger than Bhutan by chance.

We can use bootstrapping to calculate the p-value. First, we estimate the underlying distribution of the null hypothesis underlying distribution, by making a probability mass function from all of our samples from Nepal and all of our samples from Bhutan.

```
def pvalue_bootstrap(bhutan_sample, nepal_sample):
    N = size of the bhutan_sample
    M = size of the nepal_sample
    universal_sample = combine bhutan_samples and nepal_samples
    universal_pmf = estimate the underlying pmf of the universalSample
    count = 0
    observed_difference = mean(nepal_sample) - mean(bhutan_sample)
    repeat 10,000 times:
        bhutan_resample = draw N new samples from the universalPmf
        nepal_resample = draw M new samples from the universalPmf
        mu_bhutan = sample mean of the bhutanResample
        mu_nepal = sample mean of the nepalResample
        mean_difference = |muNepal - muBhutan|
        if mean_difference > observed_difference:
            count += 1
    pvalue = count / 10,000
```

This is particularly nice because nowhere did we have to make an assumption about a parametric distribution that our samples came from (ie we never had to claim that happiness is gaussian). You might have heard of a t-test. That is another way of calculating p-values, but it makes the assumption that both samples are gaussian and that they both have the same variance. In the modern context where we have reasonable computer power, bootstrapping is a more correct and versatile tool.